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E DELL'INFORMAZIONE

Executive Summary of the Thesis

Curriculum Reinforcement Learning through Teleportation: The Teleport MDP

Laurea Magistrale in Computer Science and Engineering - Ingegneria Informatica

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Academic year: 2023-2024

1. Introduction

Deep Reinforcement Learning (DRL) has emerged as a powerful approach for solving complex decision-making tasks, leading to significant advancements in robotics, game playing and autonomous systems. Despite these successes, DRL agents still struggle to match human performances in complex domains, i.e. navigating environments with sparse rewards, high-dimensional spaces and long-term credit assignment problems remains challenging.

Curriculum Learning (CL, [6]) presents a promising technique to overcome these obstacles. Inspired by the human brain's learning process, CL proposes structuring training data to gradually increase task complexity aiming to master simpler concepts before more advanced ones. An intuitive approach to designing a curriculum involves controlling the agent's starting conditions. This concept underpins *Reverse Curriculum Generation* (RCG, [2]), which begins with a goal state and generates progressively distant starting points through random walks with noise. This method effectively creates a learning path from easier to more challenging scenarios. Interestingly, we can re-frame this approach by considering early-visited states as more accessible. Unfortunately, when the reward signal is sparse, these states may offer limited learning opportunities. *Hindsight Experience Replay* (HER,[1]) addresses this challenge by enabling agents to learn from unintended outcomes. When an agent fails to reach its goal g and instead lands in state s_T, HER retrospectively replaces the intended goal with s_T. This adjustment allows *off-policy* algorithms to intentionally learn how to reach s_T, introducing an implicit curriculum.

The design of an effective curriculum necessitates a deep understanding of the relevant aspects, in particular:

- **Task Generation**: relies upon the definition of *source* problems to be learned before the final or *target* task. The quality of these intermediate steps dictates the effectiveness of the strategy.
- **Sequencing**: introduces an order over tasks to shape an optimal learning pathway. This process, arguably the most challenging one, can be approached either *manually*, or through *automated* algorithms. The development of a fully automated sequencing method is one of the main focuses of this work.

- **Transfer Learning**: involves sharing knowledge between consecutive tasks. Transfer can involve *low-level* knowledge, such as complete policies or value functions, or *high-level* knowledge, providing strategic guidance for future learning.

CL has shown intuitive appeal and empirical success, yet two key challenges have hindered its widespread adoption. First, the design of effective curricula is a complex task, often requiring domain knowledge and significant trial and error. Second, the theoretical foundations of CL remain underdeveloped, limiting our understanding of why and how curricula work, and how to design them optimally.

Our work seeks to contribute at this juncture, introducing the Teleport Markov Decision Processes (TMDPs) framework. TMDPs extend the standard Markov Decision Process (MDP) formulation by incorporating a *teleportation* mechanism. Under this setting, the agent can be relocated into any state during the middle of an episode, effectively enhancing exploration. The teleport rate τ serves as a controller for task difficulty while the following states are drawn from the teleport probability distribution ξ.

Starting with high teleport rates, the agent is frequently teleported into different states. This approach *simplifies* the task by enhancing environmental exploration while *limiting* long-term foresight reducing the effective discount factor. As the agent learns and improves, the teleport probability is gradually reduced, increasing the task's complexity and converging toward the original problem formulation. This way, the agent can exploit the knowledge acquired from simpler problems to boost learning progress on subsequent and more complex ones.

Our research makes both theoretical and practical contributions. It provides a formal mathematical framework extending the results presented in *Configurable Markov Decision Process* (CMDP, [4]) to the context of CL. It also offers a more flexible approach to curriculum design, reducing the need for domain-specific expertise. Building on this foundation, we derive practical algorithms for real-world RL scenarios. We introduce the *Teleport Model Policy Iteration* (TMPI) algorithm which, while primarily of theoretical interest, offers valuable insights into the properties of TMDPs. Then we present two practical algorithms designed to limit state visit distribution shifts during updates: *Static Teleport* (S-T) and *Dynamic Teleport* (D-T). As the name suggests, S-T algorithm employs a static scheduling; conversely, D-T monitors the impact of both policy and model updates, adjusting the teleport rate based on policy variations.

2. Preliminaries

A Markov Decision Process (MDP) is defined by the tuple $\mathcal{M} = \langle \mathcal{S}, \mathcal{A}, P, R, \gamma, \mu \rangle$ where $\mathcal{S}$ is the state space, $\mathcal{A}$ is the action space, $P(s'|s,a)$ is the transition model expressing the probability of moving from state s to state s' when action a is selected, $R(s,a) \in [0,1]$ is the reward received when choosing action a in state s, $\gamma \in [0,1]$ is the discount factor, and μ describes the initial state distribution. A policy $\pi(a|s)$ assigns a probability distribution over actions $a \in \mathcal{A}$ given the current state $s \in \mathcal{S}$. Combining a policy π with the transition model P gives rise to the state transition kernel $P^\pi(s'|s) = \sum_{a \in \mathcal{A}} \pi(a|s) P(s'|s,a)$.

The main focus of this thesis is on *Policy Gradient* (PG) methods, specifically considering *parametric policies*. In this category, a policy π_θ is uniquely identified by a parameter vector $\theta \in \Theta \subseteq \mathbb{R}^m$ and is assumed to be stochastic and differentiable w.r.t. θ. PG algorithms directly explore the policy space to solve the optimization task, coming with strengths and weaknesses. On one hand, they offer several advantages including the ability to employ gradient-based exploration using supervised learning techniques, manage continuous state and action spaces, and address partially observable environments. On the other hand, these methods introduce the *distribution shift* problem: even seemingly minor parameter adjustments can lead to significant changes in the state-action visit distribution, potentially causing performance fluctuations.

Safe Learning [3] aims to mitigate this phenomenon by introducing a dissimilarity penalization term and evaluating the worth of updates through an advantage measure. Despite the robustness of the mathematical framework, estimating these quantities in real-world applications can be challenging and may significantly increase sample complexity.

Trust Region (TR) methods pursue a similar objective but employ a slightly different strat-

egy. The idea is to constrain policy updates within a *trust region* to ensure stable learning. *Proximal Policy Optimization* (PPO, [5]) has gained prominence in the field due to its remarkable performance and practical applicability, despite—or perhaps because of—its relatively simple implementation.

3. Teleport MDP

Definition 3.1 (Teleport MDP). *A Teleport MDP (TMDP) is defined by the tuple* $\mathcal{T} = \langle \mathcal{S}, \mathcal{A}, P, R, \gamma, \mu, \tau, \xi \rangle$ *where* $\langle \mathcal{S}, \mathcal{A}, P, R, \gamma, \mu \rangle$ *is an MDP and* $\tau \in [0, 1)$ *is the teleport rate specifying the frequency at which teleport occurs and* ξ *is the state teleport probability distribution from where next states can be sampled.*

A TMDP introduces a competition between two probability distributions to govern transition dynamics, resulting in a new transition model function: $P_\tau(s'|s,a) = (1-\tau)P(s'|s,a) + \tau\xi(s')$. Like (CMDP, [4]), we must evaluate the impact of updating the model-policy pair (P_τ, π), resulting in the following performance measure:

$$J^\pi_{\mu,\tau} = \frac{1}{1-\gamma} \sum_{s\in\mathcal{S}} d^\pi_{\mu,\tau}(s) \sum_{a\in\mathcal{A}} \pi(a|s) R_\tau(s,a), \quad (1)$$

where $d^\pi_{\mu,\tau}$ identifies the γ-discounted state visit distribution recursively defined as follows:

$$d^\pi_{\mu,\tau}(s) = (1-\gamma)\mu(s) + \gamma \sum_{s'\in\mathcal{S}} d^\pi_{\mu,\tau}(s') P^\pi_\tau(s'|s). \quad (2)$$

Considering the expectation over the policy π we get the γ-discounted state-action distribution $\delta^\pi_{\mu,\tau}(s,a) = \pi(a|s) d^\pi_{\mu,\tau}(s)$. The *action-value function* is defined as the expected return when choosing action a in state s:

$$Q^\pi_\tau(s,a) = R_\tau(s,a) + \gamma \sum_{s'\in\mathcal{S}} P_\tau(s'|s,a) V^\pi_\tau(s), \quad (3)$$

whereas, to learn the transition model, we introduce the *state-action-next state value function*, U^π_τ, expressing the expected return when taking action a in state s leads into the next state s':

$$U^\pi_\tau(s,a,s') = R(s,a,s') + \gamma V^\pi_\tau(s'), \quad (4)$$

where V^π_τ is the *value function*. Furthermore, we define the *policy advantage function* $A^\pi_\tau(s,a) = Q^\pi_\tau(s,a) - V^\pi_\tau(s)$ to measure the benefits of action a in state s and the *model advantage function* $A^\pi_\tau(s,a,s') = U^\pi_\tau(s,a,s') - Q^\pi_\tau(s,a)$ to evaluate the quality of next state s'. Extending these definitions to quantify the *one-step* performance improvement when moving from a policy π to a new one π' or when changing τ to τ', we present the *relative advantage functions* [3]:

$$A^{\tau,\pi'}_{\tau,\pi}(s) = \sum_{a\in\mathcal{A}} \pi'(a|s) A^\pi_\tau(s,a),$$
$$A^{\tau',\pi}_{\tau,\pi}(s,a) = \sum_{s'\in\mathcal{S}} P_{\tau'}(s'|s,a) A^\pi_\tau(s,a,s'),$$

and the associated expectation under the γ-discounted state visit distributions:

$$\mathbb{A}^{\pi',\tau}_{\pi,\mu} = \sum_{s\in\mathcal{S}} d^\pi_{\mu,\tau}(s) A^{\tau,\pi'}_{\tau,\pi}(s),$$
$$\mathbb{A}^{\tau',\pi}_{\tau,\mu} = \sum_{s\in\mathcal{S}} \sum_{a\in\mathcal{A}} \delta^\pi_{\mu,\tau}(s,a) A^{\tau',\pi}_{\tau,\pi}(s,a) = (\tau-\tau')\mathbb{A}^{\xi,\pi}_{P,\tau,\mu},$$
$$\mathbb{A}^{\xi,\pi}_{P,\tau,\mu} = \sum_{s} \sum_{a} \delta^\pi_{\mu,\tau}(s,a) . \sum_{s'} \Big(P(s'|s,a) - \xi(s')\Big) U^\pi_\tau(s,a,s').$$

Besides defining the decoupled advantage terms, to consider the combined benefits of a model-policy update we introduce the *kernel advantage function* $\widetilde{A}^\pi_\tau(s,a,s') = U^\pi_\tau(s,a,s') - V^\pi_\tau(s)$, the *coupled relative advantage function*

$$A^{\tau',\pi'}_{\tau,\pi}(s) = \sum_{s'\in\mathcal{S}} \sum_{a\in\mathcal{A}} \pi'(a|s) P_{\tau'}(s'|s,a) \widetilde{A}^\pi_\tau(s,a,s'), \quad (5)$$

and the expectation under the γ-discounted distribution $\mathbb{A}^{\tau',\pi'}_{\tau,\pi,\mu} = \sum_{s\in\mathcal{S}} d^\pi_{\mu,\tau}(s) A^{\tau',\pi'}_{\tau,\pi}(s)$.

Although CMDP shares similarities with TMDP, their primary focus diverges. While the former aims to find the optimal model-policy pair $(P^\star, \pi^\star)$ for maximum performance, TMDPs are designed to generate a curriculum. To better understand this concept, let us examine how the teleport mechanism works. There are two main effects:: it enables *direct exploration* of the state space, offering finer control compared to *restart distributions* [3], and it *truncates* the current trajectory, restarting the agent from a state sampled from ξ. This truncation alters the effective discount factor $\gamma_{\text{eff}} = \gamma(1-\tau)$. By controlling τ, we can craft a curriculum that begins with higher values, resulting in a lower γ_{eff}, to promote exploration and short-term goals. As learning progresses, reducing τ ($\gamma_{\text{eff}} \to \gamma$) shifts the focus towards long-term planning. Therefore, evaluating the impact of both the model and the policy on performance is crucial to converge to $\tau = 0$.

4. Performance Improvement

In this section, our objective is to establish a lower bound for the performance gain when transitioning from the model-policy pair (P_τ, π) to $(P_{\tau'}, \pi')$.

4.1. State visit distribution bound

We first establish an upper bound on the shift of the γ-discounted state visitation distributions resulting from changing the model-policy pairs.

Proposition 4.1 (Proposition 5.1.1 from [4]). *Let (P_τ, π) and $(P_{\tau'}, \pi')$ two model-policy pairs, the L_1-norm of the difference between the γ-discounted state distributions can be upper-bounded as:*

$$\left\|d^{\pi'}_{\mu,\tau'} - d^{\pi}_{\mu,\tau}\right\|_1 \leq \frac{\gamma}{1-\gamma} D_{\mathbb{E}}^{P^{\pi'}_{\tau'}, P^{\pi}_{\tau}}, \tag{6}$$

where $D_{\mathbb{E}}^{P^{\pi'}_{\tau'}, P^{\pi}_{\tau}} := \mathbb{E}_{s\sim d^{\pi}_{\mu,\tau}} \left\|P^{\pi'}_{\tau'}(\cdot|s) - P^{\pi}_{\tau}(\cdot|s)\right\|_1$.

This proposition bounds the difference between γ-discounted state distributions using the state kernel dissimilarity. While the state kernel captures the combined effects of policy and transition model variation, practically, it is useful to separate their contributions.

Corollary 4.1. *Let (P_τ, π) and $(P_{\tau'}, \pi')$ two model-policy pairs. The L_1-norm of the difference between the γ-discounted state distributions can be upper-bounded as:*

$$\left\|d^{\pi'}_{\mu,\tau'} - d^{\pi}_{\mu,\tau}\right\|_1 \leq \frac{\gamma}{1-\gamma}(D_{\mathbb{E}}^{\pi',\pi} + |\tau - \tau'| D_{\mathbb{E}}^{P,\xi}), \tag{7}$$

where $D_{\mathbb{E}}^{\pi',\pi} := \mathbb{E}_{s\sim d^{\pi}_{\mu,\tau}} \left\|\pi'(\cdot|s) - \pi(\cdot|s)\right\|_1$ *and* $D_{\mathbb{E}}^{P,\xi} := \mathbb{E}_{(s,a)\sim \delta^{\pi}_{\mu,\tau}} \left\|P(\cdot|s,a) - \xi(\cdot)\right\|_1$.

Note. *The absolute value on $|\tau' - \tau|$ is necessary for mathematical rigour, however, in our settings, the goal is a monotonic convergence towards $\tau = 0$, thus the absolute value can be omitted without loss of generality.*

4.2. Performance improvement bound

We will now derive a series of lower bounds on performance improvement. Leveraging (Equation 5) we can establish the following lower bound on performance improvement.

Theorem 4.1 (Theorem 5.1.1 from [4]). *The performance improvement of the model-policy pair (P_τ, π) over $(P_{\tau'}, \pi')$ is given by:*

$$J^{\pi'}_{\mu,\tau'} - J^{\pi}_{\mu,\tau} = \frac{1}{1-\gamma} \sum_{s\in\mathcal{S}} d^{\pi'}_{\mu,\tau'}(s) A^{\tau',\pi'}_{\tau,\pi}(s). \tag{8}$$

The nonlinear relation between $d^{\pi'}_{\mu,\tau'}$ and the new model-policy pair $(P_{\tau'}, \pi')$ prevents a direct application of this theorem. Therefore, we intend to establish a lower bound that relies only on the decoupled contribution of the current model-policy pair (P_τ, π).

Theorem 4.2 (Teleport Bound). *The performance improvement of the model-policy pair $(P_{\tau'}, \pi')$ over (P_τ, π), can be lower bounded as:*

$$\underbrace{J^{\pi'}_{\mu,\tau'} - J^{\pi}_{\mu,\tau}}_{\substack{\textit{performance}\\ \textit{improvement}}} \geq \underbrace{\frac{\mathbb{A}^{\pi',\tau}_{\pi,\mu} + (\tau - \tau')\mathbb{A}^{\xi,\pi}_{P,\tau,\mu}}{1-\gamma}}_{\textit{advantage}} - \underbrace{\frac{\gamma \Delta U^{\pi}_{\tau} D}{2(1-\gamma)^2}}_{\substack{\textit{dissimilarity}\\ \textit{penalization}}}$$

where D is a dissimilarity term defined as:

$$D = D_{\mathbb{E}}^{\pi',\pi}(D_{\infty}^{\pi',\pi} + |\tau - \tau'| D_{\infty}^{P,\xi}) + \\ + (\tau - \tau') D_{\mathbb{E}}^{P,\xi}(D_{\infty}^{\pi',\pi} + \gamma|\tau - \tau'| D_{\infty}^{P,\xi}).$$

and $\Delta U^{\pi}_{\tau} = \sup_{\substack{s,s',\overline{s},\overline{s}'\in\mathcal{S},\\ a,a'\in\mathcal{A}}} \left|U^{\pi}_{\tau}(s',a',\overline{s}') - U^{\pi}_{\tau}(s,a,\overline{s})\right|$

5. TMDP-based Curricula

In this section, we present the three TMDP-based curriculum algorithms: *Teleport Model Policy Iteration* (TMPI), *Static-Teleport* (S-T) and *Dynamic-Teleport* (D-T). As anticipated, curriculum design encompasses task generation, sequencing, and transfer learning. We adopt a low-level transfer approach, initializing the policy and value functions of the new task $\tau^{(t+1)}$ using those learned with $\tau^{(t)}$. The remaining phases are jointly addressed by determining *when* and *how* to update the teleport rate τ.

5.1. Teleport Model Policy Iteration

The general idea of TMPI is to start with higher values of τ, whose initial value becomes a hyperparameter, and then iteratively update both the policy π and the teleport rate τ with the rules:

$$\pi' = \alpha^\star \overline{\pi} + (1-\alpha^\star)\pi, \qquad \alpha^\star \in [0,1],$$
$$P_{\tau'} = (1-\tau^\star)P + \tau^\star \xi, \qquad \tau^\star \in [0,\tau],$$

where $\overline{\pi}$ is the target policy, $\alpha^\star$ and $\tau^\star$ are the values that jointly optimize Theorem 4.2.

Note. *The model update is a consequence of updating the teleport rate $\tau' \leftarrow \tau^\star$, furthermore, we will refer to the Teleport Bound as $B(\alpha, \tau')$.*

The following theorem presents a significant result concerning the lower bound optimization.

Theorem 5.1. *For any policy $\overline{\pi} \in \Pi$ and teleport transition imodel $P_\tau \in \mathcal{P}$, the lower bound is optimized for:*

$$\alpha^\star, \tau^\star = \arg\max_{\alpha,\tau'} \{B(\alpha,\tau') : (\alpha,\tau') \in \mathcal{V}\},$$

where $\mathcal{V} = \{(\overline{\alpha}_\tau, \tau, (0, \overline{\tau}_0), (\overline{\alpha}_0, 0), (1, \overline{\tau}_1)\}$. The candidate coefficients are presented in Table 1.

Table 1: Candidate optimal pairs for the teleport bound. All values are clipped within the range $[0, 1]$.

$\overline{\alpha}_\tau = \frac{(1-\gamma)\mathbb{A}^{\tau,\pi'}_{\tau,\pi,\mu}}{\gamma \Delta U^\pi_\tau D^{\pi',\pi}_{\mathbb{E}} D^{\pi',\pi}_{\infty}}$	$\overline{\alpha} = 0$	$\overline{\alpha}_0 = \overline{\alpha}_\tau - \frac{\tau}{2}\left(\frac{D^{P,\xi}_{\mathbb{E}}}{D^{\pi',\pi}_{\mathbb{E}}} + \frac{D^{P,\xi}_{\infty}}{D^{\pi',\pi}_{\infty}}\right)$	$\overline{\alpha} = 1$
$\overline{\tau}' = \tau$	$\overline{\tau}'_0 = \tau - \frac{(1-\gamma)\mathbb{A}^{\xi,\pi}_{P,\tau,\mu}}{\gamma^2 \Delta U^\pi_\tau D^{P,\xi}_{\mathbb{E}} D^{P,\xi}_{\infty}}$	$\tau' = 0$	$\overline{\tau}'_1 = \overline{\tau}'_0 + \frac{1}{2\gamma}\left(\frac{D^{\pi',\pi}_{\mathbb{E}}}{D^{P,\xi}_{\mathbb{E}}} + \frac{D^{\pi',\pi}_{\infty}}{D^{P,\xi}_{\infty}}\right)$

The theorem specifies that the optimal pair $(\alpha^\star, \tau^\star)$ lies on the boundaries, defined over the space $[0, 1] \times [0, \tau]$. Consequently, either the policy or the teleport rate is updated while the other remains fixed, or one is updated and the other is set to target.

Algorithm 1 outlines the general structure of TMPI. The main difference from traditional reinforcement learning algorithms lies in the sampling loop, where the agent's next state is determined not only by the environment's transition dynamics but also by the teleport rate τ and the associated probability distribution ξ. The easiest way to implement this behaviour involves truncating the episode upon teleportation and restarting the agent from a state sampled from ξ, however, more sophisticated implementations provide greater flexibility.

Algorithm 1 Teleport Model Policy Iteration (TMPI)

Require: Initial policy $\pi^{(0)}$, teleport rate $\tau^{(0)}$, batch size N, epochs E.
for t=0,1,2... until convergence **do**
  for N trajectories **do**
    while $\neg$*terminated* $\vee$ $\neg$*teleport* **do**
      Sample $p \sim U(0, 1)$
      if $p \leq \tau^{(t)}$ **then**
        $s' \sim \xi$
        teleport $= True$
      else
        Sample $a \sim \pi^{(t)}(\cdot|s)$
        Observe reward r, next state s'
        ...
      end if
    end while
  end for
  Learn $\overline{\pi}^{(t)}$
  Compute $\mathbb{A}^{\pi',\tau}_{\pi,\mu}$, $\mathbb{A}^{\xi,\pi}_{P,\tau,\mu}$, ΔU^π_τ and D
  Compute candidate pairs:
$$\mathcal{V}^{(t)} = \{(\overline{\alpha}^{(t)}_\tau, \tau^{(t)}), (0, \overline{\tau}^{(t)}_0), (\overline{\alpha}^{(t)}_0, 0), (1, \overline{\tau}^{(t)}_1)\}$$
  Compute optimal pairs:
$$\alpha^\star, \tau^\star = \arg\max_{\alpha,\tau}\{B(\alpha, \tau) : (\alpha, \tau) \in \mathcal{V}^{(t)}\}$$
  $\pi^{(t+1)} = \alpha^\star \overline{\pi}^{(t)} + (1 - \alpha^\star)\pi^{(t)}$
  $\tau^{(t+1)} \leftarrow \tau^\star$
end for

TMPI offers valuable theoretical insights but faces practical limitations when the model dynamics are unknown or when dealing with negative rewards. The dissimilarity penalization terms can be effectively upper-bounded, still, the overall quality of the solution depends on the accuracy of the advantage term estimates. A potential solution is to develop a sampling technique similar to the one employed by [3], however, the estimation of the model advantage necessitates crossing into the realm of *model-based* approaches, which falls outside the scope of this thesis. Taking inspiration from Trust Region methods we now consider how to pursue a similar objective in a context where both the policy and the model are updated.

5.2. Static and Dynamic Teleport

Starting from Corollary 4.1, and considering the *stocasticity* of P and ξ we can devise a *model-free* version of the lower bound on the difference of the state visit distributions.

Corollary 5.1. *Let (P_τ, π) and $(P_{\tau'}, \pi')$ two model-policy pairs. The L_1-norm of the difference between the γ-discounted state distributions can be upper-bounded as:*

$$\left\|d^{\pi'}_{\mu,\tau'} - d^{\pi}_{\mu,\tau}\right\|_1 \leq \frac{\gamma}{1-\gamma}(D^{\pi',\pi}_{\mathbb{E}} + 2|\tau - \tau'|), \quad (9)$$
$$\leq \epsilon_\pi + \epsilon_\tau = \epsilon$$

Algorithm 2 Static Teleport (S-T)

Require: Initial policy $\pi^{(0)}$, teleport rate $\tau^{(0)}$, learning rate α, teleport update rate β, batch size N, epochs E, total time steps T.
Compute the number of updates: $n = T\beta$
Compute $\epsilon_\tau = \frac{2\gamma\tau}{n(1-\gamma)}$
for t=0,1,..., T **do**
  Collect N trajectories
  Learn $\pi^{(t+1)}$
  $\tau^{(t+1)} = \tau^{(t)} - \epsilon_\tau \frac{(1-\gamma)}{2\gamma}$
end for

Considering first the effect of updating only the model, i.e. $\pi' = \pi$ we can state the following:

Theorem 5.2. *Let π a fixed policy, $d^{\pi}_{\mu,\tau}$ the γ-discounted state visit distribution and $\epsilon_\tau \in \mathbb{R}^+$ a model shift threshold, the following update rule:*

$$\tau^{(t+1)} \leftarrow \tau^{(t)} - \epsilon_\tau \frac{(1-\gamma)}{2\gamma}, \tag{10}$$

guarantees that:

(i) the L_1 norm of the difference between the γ-discounted state visit distribution of every single update of the teleport rate τ under the same policy π is upper bounded as:

$$\|d^{\pi}_{\mu,\tau^{(t+1)}} - d^{\pi}_{\mu,\tau^{(t)}}\|_1 \leq \frac{2\gamma|\tau - \tau'|}{(1-\gamma)} \leq \epsilon_\tau.$$

(ii) The teleport rate τ converges to 0 in a fixed number of updates $n = \left\lceil \frac{2\gamma\tau}{\epsilon_\tau(1-\gamma)} \right\rceil$.

As a direct consequence, we can compute the threshold ϵ_τ that ensures a convergence from $\tau^{(0)} = \tau$ to $\tau^{(n)} = 0$ as follows: $\epsilon_\tau \approx \frac{2\gamma\tau}{n(1-\gamma)}$.

This theorem provides the theoretical foundation for implementing a practical and efficient static curriculum, as outlined in Algorithm 2. Notably, *Static-Teleport* (S-T) allows us to control the learning process without prior knowledge of the transition dynamics. By ensuring that the γ-discounted state visit distribution does not change more than a given threshold ϵ_τ, we can arbitrarily limit the distribution shift of each update. This constraint promotes a more stable learning process, following the clipping approach adopted in (PPO, [5]).

To develop a dynamic technique that adapts to the agent's learning progress, we must consider the policy contribution too. Dynamic Teleport (D-T, Algorithm 3) adjusts the teleport rate in response to the agent's learning progress, balancing distribution shifts caused by both policy and model changes. Given a total allowed shift ϵ, the algorithm first computes ϵ_π to quantify the distribution shift associated with the policy update. Trust Region methods are particularly well-suited for policy learning in this context, as they provide a way to control this step size. Subsequently, ϵ_τ is computed as the difference between the total allowed shift ϵ and the policy-induced shift ϵ_π, effectively representing the remaining *budget* for model updates. If a model update is allowed ($\epsilon_\tau > 0$), the actual step size is further constrained to prevent excessive changes. Informally, when the policy undergoes significant changes, reflecting that the agent is acquiring new knowledge, the teleport rate update is reduced or potentially skipped. Conversely, when the policy stabilizes, it suggests that the agent might be capable of handling increased task complexity, thus warranting a reduction in the teleport rate. It is worth mentioning that both Algorithm 2 and Algorithm 3 present a general structure that can be used to *augment* existing RL algorithms to handle *teleport-based* curricula.

Algorithm 3 Dynamic Teleport (D-T)

Require: Initial policy $\pi^{(0)}$, teleport rate $\tau^{(0)}$, learning rate α, state distribution shift ϵ, maximum model shift $\epsilon_{\tau_{\max}}$, batch size N, epochs E.

```
for t=0,1,... until convergence do
    Collect N trajectories
    Learn π^(t+1)
    Compute ε_π = γ/(1−γ) D_∞^{π',π}
    Compute ε_τ = ε − ε_π
    if ε_τ > 0 and τ^(t) > 0 then
        τ^(t+1) = τ^(t) − min(ε_τ, ε_τmax) (1−γ)/(2γ)
    end if
end for
```

6. Experimental Evaluation

Our empirical evaluation, assessed the performance of two widely used RL algorithms, PPO and Q-learning, by comparing their vanilla implementations against TMDP-based curricula. This study was conducted in a tabular setting with discrete state and action spaces, specifically focusing on the *Frozen Lake* and *River Swim* environments, known for their sparse reward and exploration challenges.

In the *Frozen Lake*[1] domain, our initial experiments demonstrated that teleport-based algorithms consistently reached the goal state, whereas their vanilla counterparts failed. To ensure a fair comparison, we introduced a reward-shaping technique based on the *Manhattan* distance. Each state was assigned to a discrete bin with linear rewards ranging from $[-1, 0]$, while the goal state received a 1 reward.

The results on average return (Figure 1a) highlight that PPO-based curricula outperform all

[1] Frozen Lake is a classic grid world environment with sparse rewards where the agent must navigate from a start state toward a goal state.

other algorithms, with D-TPPO achieving the best overall performance. In contrast, vanilla implementation struggled the most, scoring the lowest and failing to learn the original task. Figure (1b) provides a perspective on the teleport rate updates. Focusing on the original setting, the two static teleport implementations (S-TPPO, S-TQ) follow a linear scheduler. In contrast, both TMPI and D-TPPO adapt the teleport rate based on learning performance. TMPI acquired substantial knowledge and learned the optimal policy before adjusting τ. On the other hand, D-TPPO began updates earlier, temporarily pausing when facing higher uncertainty and resuming adjustments later.

Moving to the *River Swim*[2] environment, the goal was to evaluate the performance of teleport-based curricula by comparing a uniform restart distribution against the original initial distribution, where the agent starts deterministically in the leftmost state.

Figure (1c) presents the average return of *River Swim* experiments, revealing similar trends to those observed in Frozen Lake. Once again, the TMDP-based PPO implementation demonstrates superior performance, while at the opposite end of the performance spectrum, we find the two vanilla implementations, PPO and Q-learning. S-TQ exhibits consistent performance across both settings, with a slight reduction in variance observed in the original setting. On the other hand, TMPI displays a notable improvement in the original setting compared to the uniform restart scenario. Both behaviours could be linked to excessive exploration when combining the two approaches, as their similar effects might lead to overlapping strategies.

In conclusion, our results underscore the potential of the TMDP framework to enhance curriculum learning, paving the way for future works.

7. Conclusions

Our research provides both theoretical and practical contributions. Extending the traditional MDP formulation we have introduced a novel approach for Curriculum Learning in RL: the Teleport Markov Decision Process (TMDP). Building upon results presented in (CMDP, [4]), we have established a solid mathematical foundation, deriving lower bounds for performance improvement. This led to the definition of the *Teleport Model Policy Iteration* (TMPI) algorithm, which offers valuable insights into the properties of our framework. Then, to overcome its practical limitations, we derived two real-world algorithms *Static-Teleport* (S-T) and *Dynamic-Teleport* (D-T) enabling the integration of teleport-based curricula into standard RL algorithms. Finally, we presented empirical evidence that shows the effectiveness of teleport-based curricula on tested domains.

[2] River Swim environment consists of a linear chain of states with the agent navigating against a strong current.

References

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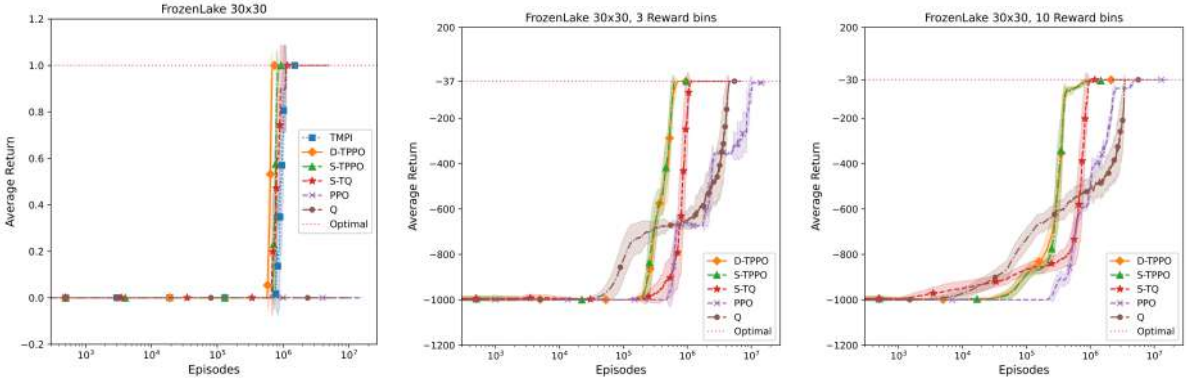
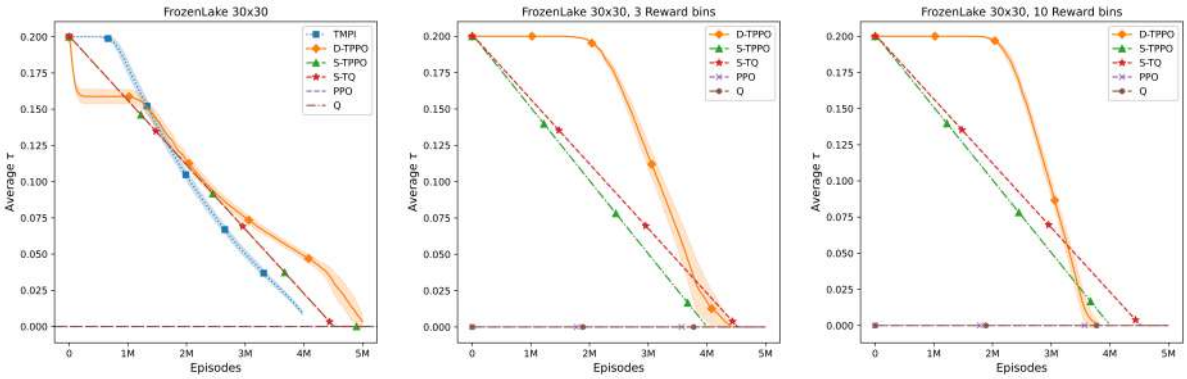
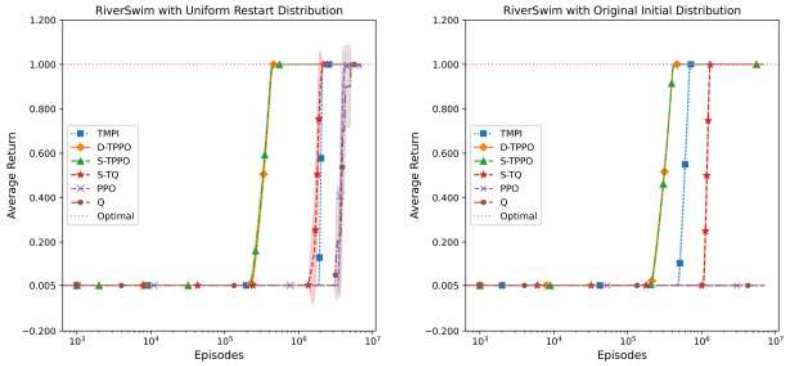


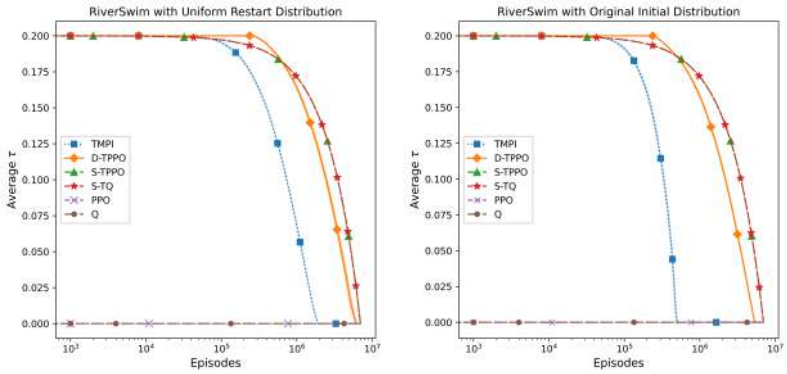
(a) Average return for Frozen Lake 30×30 with 0, 3, and 10 reward bins.



(b) Average teleport rate for Frozen Lake 30×30 with 0, 3, and 10 reward bins.



(c) Average return for River Swim 70 with (left) and without (right) uniform restart distribution.



(d) Average teleport rate for River Swim 70 with (left) and without (right) uniform restart distribution.

Figure 1: Experiment results for Frozen Lake and River Swim environments. Average results on 10 runs with random seeds, 95% C.I.